A

Major Project Report on

"PROJECT PILOT"

Submitted in partial fulfilment of the requirement of the University of

Mumbai for the Degree of **Bachelor of Engineering**

(Computer Engineering)

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CERTIFICATE

This is to certify that project entitled **"PROJECT PILOT"** is a Bonafide work of **Mr. Anuj Dighe, Mr. Mayur Kumawat, Ms. Chaitali Dhaije, Mr. Om Waghmare** submitted to the University of Mumbai in partial fulfilment the requirement for the award of degree of "**Bachelor of Engineering**" in "**Computer Engineering**" in "**Computer Engineering**".

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Major Project Approval

This Major Project entitled “PROJECT PILOT” by **Mr. Anuj Dighe, Mr. Mayur Kumawat, Ms. Chaitali Dhaije, Mr. Om Waghmare,** is approved for the degree of Bachelor of Engineering in Computer Engineering.

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Place: DRIEMS, Neral

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

Project management in academic environments often lacks a structured digital platform that bridges the gap between students and mentors. ProjectPilot is a project management application developed using the MERN (MongoDB, Express.js, React, Node.js) stack, designed specifically for academic project tracking and collaboration. The system provides dedicated dashboards for students, professors (mentors), and an administrator responsible for mentor allocation. Students can create and manage projects through features such as a Kanban board for task tracking, group formation, report submissions, and receiving structured feedback from mentors.

Professors are enabled with tools to monitor project progress, review submissions, and guide students effectively. The platform also supports project notifications, name submissions, and real-time updates to ensure seamless communication. By integrating collaboration, tracking, and mentorship into one unified system, ProjectPilot enhances project organization, transparency, and efficiency in academic settings. This project demonstrates the application of modern full-stack web technologies to deliver a scalable, interactive, and user-friendly solution for student–mentor project management.

List of Abbreviations

JWT	JSON Web Token
APIs	Application Programming Interfaces
UI/UX	User Interface/User Experience

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INTRODUCTION

1.1 Introduction:

ProjectPilot is an innovative **project management and collaboration platform** developed using the **MERN stack (MongoDB, Express.js, React.js, and Node.js)**. The platform is specifically designed to **streamline academic project tracking, supervision, and communication** among students, mentors, and administrators within educational institutions.

In most colleges and universities, managing academic projects can become complex and time-consuming due to the involvement of multiple stakeholders — students, faculty guides, internal evaluators, and department coordinators. ProjectPilot bridges this gap by offering an integrated, user-friendly system that centralizes all aspects of project management.

The platform offers **dedicated dashboards** tailored to different roles:

- **Students** can create projects, form groups, assign tasks, maintain progress through a **Kanban board**, and submit their reports and documentation digitally.
- **Professors or mentors** can review submissions, provide structured feedback, monitor real-time progress, and guide their assigned groups effectively.
- **Administrators** can manage project approvals, mentor allocations, and overall progress tracking of multiple batches or departments.

The system enhances transparency, accountability, and productivity by automating repetitive administrative tasks and reducing dependency on manual coordination through emails or paperwork. Additionally, **real-time notifications and updates** ensure that all participants remain informed about deadlines, status changes, and mentor feedback.

By integrating these features into a unified digital platform, ProjectPilot aims to revolutionize the way academic institutions manage projects, making the entire process **organized, collaborative, and outcome-driven**.

1.2 Motivation:

Managing academic projects is often challenging due to the lack of a centralized platform that connects students, mentors, and administrators effectively. Traditional methods of project tracking through manual updates, emails, or scattered tools lead to inefficiency, miscommunication, and delays in progress. This motivated the development of ProjectPilot, a structured, technology-driven solution that streamlines project management in academic settings. The goal is to provide students with an organized way to create, track, and collaborate on projects, while enabling mentors to monitor progress and provide timely guidance, thereby improving overall efficiency, accountability, and communication.

1.3 Problem statement & Objectives Problem Statement:

In academic environments, managing student projects is often a challenging and fragmented process due to the **absence of a centralized digital management system**. Students and mentors typically depend on multiple, disconnected tools such as email, spreadsheets, and messaging apps for project updates, file sharing, and communication. This decentralized approach results in poor coordination, missed deadlines, version conflicts, and inadequate feedback tracking.

Existing project management tools available in the market (such as Trello, Jira, or Asana) are either **too generic** or **lack academic-specific functionalities** like mentor allocation, structured feedback modules, and multi-level approval workflows. Furthermore, these platforms are not customized to handle the hierarchical structure of educational institutions, where professors supervise multiple students and administrators oversee academic compliance.

To address these issues, ProjectPilot proposes a **dedicated academic project management platform** that consolidates all essential functionalities — project creation, group management, mentor supervision, progress tracking, and communication — into a single, integrated web application.

Objectives:

 Provide separate dashboards for students, professors (mentors), and administrators.

 Implement a Kanban board for task creation and project tracking.

 Enable group formation, project submissions, and feedback sharing.

 Provide real-time notifications to ensure effective communication.

1.4 Organization of the Report

This report is structured into several chapters to present the project in a systematic and logical manner:

- **Chapter 1:** Provides the introduction, background, motivation, problem statement, and objectives of the project, offering a clear understanding of the purpose and scope of ProjectPilot.
- **Chapter 2:** Presents a **literature survey** of existing project management applications and their limitations, highlighting the research gap that this system addresses.
- **Chapter 3:** Describes the **proposed system** in detail, including system architecture, data flow diagrams, functional components, and technology stack used.
- **Chapter 4:** Focuses on **implementation, experimentation, and results**, showcasing how the system was developed, tested, and validated with real use cases.
- **Chapter 5:** Summarizes the project with a **conclusion** and discusses possible **future enhancements** such as AI-based analytics, automated grading, or cloud deployment.
- **Chapter 6:** Lists **references, research papers, and related resources** that were used during the design and development of this project.

LITERATURE SURVEY

2.1. Survey of Existing Sytem

In recent years, numerous project management platforms have been developed to facilitate collaboration, task tracking, and productivity in organizational and educational settings. Popular tools such as **Trello**, **Asana**, **Jira**, and **Microsoft Teams** are widely adopted in the corporate world for managing projects, assigning responsibilities, and maintaining transparency among team members. However, despite their effectiveness in professional environments, these tools often fall short when applied to **academic project management**.

In an academic context, the project lifecycle involves multiple roles — students, mentors (professors), and administrators — each requiring different permissions, workflows, and tracking methods. Traditional tools like Trello and Asana primarily cater to general team-based project management and lack the academic-specific features such as **automatic mentor allocation**, **individual student dashboards**, **report submission modules**, and **feedback management** systems. Similarly, while Jira is excellent for software development workflows, it requires complex setup and technical knowledge, making it unsuitable for students or institutions seeking a lightweight, academic-focused solution.

Microsoft Teams, though commonly used in educational environments for communication and file sharing, does not offer dedicated project management features such as Kanban boards or structured progress tracking. It focuses more on chat, meetings, and document collaboration rather than systematic project lifecycle management.

The proposed system, **ProjectPilot**, has been designed to fill this gap by integrating all the essential features required for academic project supervision and collaboration. It combines the simplicity of task management tools like Trello with the structural depth required for mentor–student coordination. The following table provides a comparative overview of existing systems versus the proposed ProjectPilot system.

Feature	Trello	Asana	Jira	Microsoft Teams	ProjectPilot (Proposed)
Student Dashboard	✅	✅	✅	❌	✅
Mentor Dashboard	❌	❌	❌	❌	✅
Automatic Mentor Allocation	❌	❌	❌	❌	✅
Kanban Board & Project Tracking	✅	✅	✅	❌	✅
Reports & Feedback Module	❌	✅	✅	❌	✅
Group Collaboration	✅	✅	✅	✅	✅
Notifications & Alerts	✅	✅	✅	✅	✅

From the table, it is evident that while existing tools provide some overlapping features such as collaboration, task tracking, and notifications, none of them deliver **a comprehensive academic project management ecosystem**. ProjectPilot addresses this deficiency by introducing role-based dashboards, automated mentor allocation, and structured reporting — all of which are specifically tailored for academic use.

2.2. Limitations of Existing Systems or Research Gap

Despite the availability of advanced project management software, most existing solutions are **general-purpose tools** that do not fully cater to the unique requirements of academic project supervision. These platforms often lack several key functionalities needed in educational environments, such as:

- **Dedicated Dashboards for Students and Mentors:** Existing tools provide generic workspaces but do not differentiate between academic roles. Students, mentors, and administrators need different levels of access, visibility, and controls. For example, a mentor should be able to view progress across multiple student groups, while a student should see only their assigned project.
- **Automatic Mentor Allocation:** No mainstream tool offers the ability to automatically allocate mentors to student projects based on predefined criteria such as department, specialization, or workload. This process remains largely manual and time-consuming in most institutions.
- **Structured Reports and Feedback:** Tools like Trello and Asana support attachments and comments but lack a formalized

report submission and feedback system that tracks document versions and mentor evaluations.

- **Academic Progress Tracking:**
  Corporate tools are designed around deadlines and sprints, not academic milestones such as synopsis submission, mid-term evaluations, and final report reviews. As a result, progress visualization in academic contexts becomes difficult.
- **User Accessibility and Complexity:**
  Applications like Jira, though feature-rich, are complex for non-technical users. They require prior setup, configuration, and sometimes paid plans, making them impractical for students and educators in a typical college setting.

These limitations highlight a significant **research gap** in the area of academic project management systems. There is a growing need for a **custom-built platform** that combines ease of use with functionality tailored to educational workflows. ProjectPilot was conceptualized to bridge this gap — it provides role-specific dashboards, automated processes, and seamless communication tools that address the shortcomings of traditional systems.

By studying the limitations of existing tools and incorporating targeted improvements, ProjectPilot contributes to the advancement of digital project management in academia. It not only supports task and report management but also encourages collaboration, accountability, and real-time mentorship.

2.3 Major Project Contribution

The **ProjectPilot** system introduces a set of innovative features designed specifically for **academic environments**, ensuring a smooth and transparent project management process. The key contributions of the proposed system are summarized as follows:

- **Student and Mentor Dashboards:** Separate and intuitive dashboards are provided for students, mentors, and administrators. Students can create projects, view assigned tasks, and track progress, while mentors can monitor multiple projects, provide timely feedback, and evaluate submissions.
- **Project Creation and Tracking:** ProjectPilot integrates a **Kanban-style project tracking system**, allowing users to visualize workflows and monitor task completion stages. This ensures clarity, organization, and accountability throughout the project lifecycle.
- **Report Submission and Feedback System:** A built-in mechanism enables students to upload project reports and receive structured mentor feedback in real time. This promotes continuous improvement and eliminates the delays associated with offline submissions.
- **Group Collaboration:** The system allows students to form project groups, communicate internally, and share ideas, resources, and updates. This collaborative approach fosters teamwork and effective communication among peers.
- **Notifications and Alerts:** ProjectPilot offers **real-time notifications** and alerts to inform users about task updates, mentor feedback, submission deadlines, and announcements. This feature ensures that all participants remain actively engaged and informed.
- **Role-Based Access Control (RBAC):** The system enforces access permissions based on user roles, ensuring that sensitive data such as grades, evaluations, and feedback are securely managed.
- **Scalability and Extensibility:** The system is designed using the **MERN stack**, making it scalable for deployment across multiple departments or institutions. Additional features like automated grading or AI-based analytics can be integrated in future versions.

PROPOSED SYSTEM

3.1 Introduction

The proposed system, **ProjectPilot**, is an intelligent, web-based project management platform designed to simplify and enhance the process of **academic project supervision**. It bridges the communication and coordination gap between **students, mentors, and administrators** by integrating all project-related activities into one centralized system.

Built using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**, ProjectPilot delivers a seamless and responsive experience for users through a modern, full-stack JavaScript framework. The system offers **dedicated dashboards**, **Kanban-style task management**, **real-time progress tracking**, **report submission and feedback modules**, and **notification systems**, all of which ensure smooth workflow and collaboration.

Unlike generic project management tools, ProjectPilot is designed specifically for the **academic environment**, where multiple projects run simultaneously under the supervision of faculty mentors. It supports multiple user roles (student, mentor, and administrator), each with specific permissions and responsibilities. The platform emphasizes usability, data security, and transparency, providing every stakeholder with clear insights into the project's progress.

The system's key features include:

- Role-based dashboards tailored to user type.
- Secure authentication using JSON Web Tokens (JWT).
- Real-time notifications and updates.
- Task and project visualization using Kanban boards.
- Cloud-based storage and retrieval of project and user data.
- RESTful API-based communication between frontend and backend.

By providing a **comprehensive digital ecosystem**, ProjectPilot minimizes the manual workload involved in academic project tracking, enhances mentor–student interaction, and promotes accountability and productivity throughout the project lifecycle.

3.2. Architecture/Framework

Frontend: React.js (for web dashboards for students, mentors, and solo users)

Backend: Node.js with Express.js

Database: MongoDB (for storing project, user, and group data)

Authentication: JWT-based authentication for secure login

APIs & Services: REST APIs for project management, real-time notifications, and feedback submission

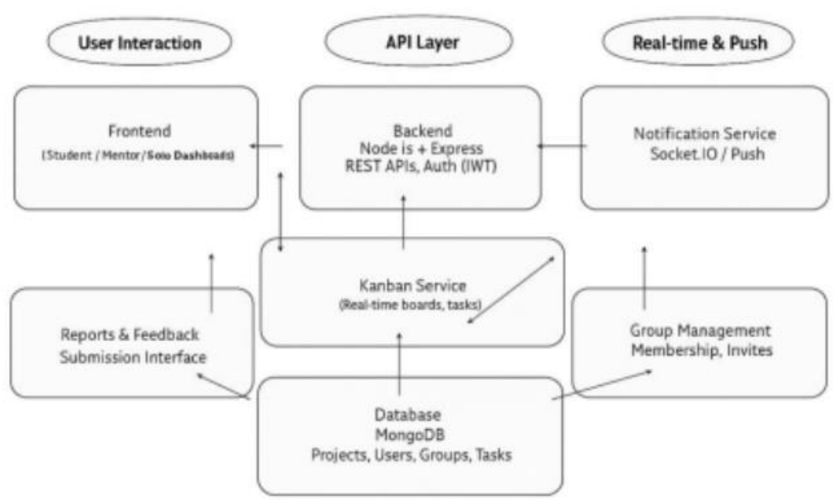

Figure: 3.2.1

3.3 Algorithm and Process Design

User Registration & Authentication (MERN Stack – MongoDB, Express, React, Node.js).

Students, professors, and solo users can register and log in securely to access their respective dashboards.

Dashboard Management.

Separate dashboards for students, professors (mentors), and solo users. Solo users can be assigned a

mentor to guide their projects.

Project & Task Management.

Students can create projects, track progress via Kanban boards, create groups, submit reports, and receive

feedback from mentors.

Notifications & Updates.

Real-time notifications for project status, feedback, group activities, and deadlines using backend event

handling.

Project Submission & Tracking.

Users can submit project names, track tasks, and manage all project-related activities efficiently within the platform.

3.4. Details of Hardware and Software:

Component	Details
Software	MERN Stack (MongoDB, Express, React, Node.js), Kanban Board Libraries, Notification Services
Hardware	Desktop/Laptop, Android/iOS devices for mobile access, Internet-enabled device

EXPERIMENT AND RESULTS

The ProjectPilot system was tested on multiple parameters to evaluate efficiency, usability, and project

management effectiveness. The experimental results are summarized as follows:

4.1 Project Creation & Tracking

The project creation and Kanban tracking system was tested under different usage conditions. The results indicate that:

 Students can create new projects and tasks within 2–4 seconds under normal network conditions.

 Task updates and status changes on the Kanban board reflect in real-time with minimal delay (≤3 seconds) on stable internet.

 Offline or slow network conditions may delay updates by up to 6 seconds, but changes synchronize

automatically once the connection is restored.

 Mentors can assign tasks and monitor progress instantly through the dashboard.

4.2 Notifications & Feedback System

The notification system for task updates, project reports, and mentor feedback was evaluated:

 Real-time notifications are delivered within 2–5 seconds on stable connections.

 Feedback submissions by mentors appear immediately on student dashboards.

 Push notifications ensure that all group members receive updates simultaneously.

 Users can manage notification preferences for projects, groups, and deadlines effectively.

4.3 User Feedback & Usability Testing

A beta version of ProjectPilot was tested by 30 students and mentors to assess usability, performance,

and overall satisfaction. Findings include:

 88% of users found the dashboard layout intuitive and easy to navigate.

 92% of students could create projects and update task status without guidance.

 80% of mentors found the feedback system efficient for reviewing submissions.

 75% of users suggested adding more customization options for group management and notifications.

 70% requested integration with calendar tools for automated deadline tracking.

CONCLUSION AND FUTURE WORK

The ProjectPilot system successfully provides an efficient project management platform with real-time task tracking, Kanban boards, and instant mentor feedback. Built using the MERN stack for full-stack functionality, the platform ensures secure, reliable, and seamless communication between students,mentors, and solo users.

The system has demonstrated high efficiency in project creation and tracking, quick updates for task status changes, and positive user feedback regarding its usability and dashboard design. By leveraging real-time database updates and notification services, users can ensure that project progress, group activities, and mentor feedback are delivered instantly, even under slow network conditions. Overall, ProjectPilot is a robust, accessible, and effective project management solution, empowering users with advanced tools for managing and collaborating on projects efficiently.

Future Work

 Calendar & Deadline Integration: Sync project deadlines and tasks with external calendar tools.

 AI-powered Task Suggestions: Recommend task prioritization and workflow optimization

based on project data.

 Mobile & Wearable Notifications: Extend notifications and task updates to smartwatches and mobile devices for instant alerts.

REFERENCES

 MERN Stack Documentation – https://mern.io/docs

• Covers MongoDB, Express, React, and Node.js used for full-stack project development.

 React Official Documentation – https://react.dev/docs

• Used for building the web front-end and student/mentor dashboards.

 Kanban Board Libraries & Guides – https://kanbanize.com/kanban-resources/getting-started/what-is-kanban

• Provides tools and guidelines for implementing task boards and project tracking.

 Notification Services Documentation – https://www.npmjs.com/package/node-notifier

• Used for sending real-time updates, alerts, and notifications to users.